Flow and AI — Research Paper

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Flow and AI in Video Games Methods and the User Experience November 20th, 2022 Airron Kennedy Instructor Robert Kennedy

Flow and AI in Video Games Video games have been around for over 30 years, and during that time they've evolved into an increasingly popular form of entertainment. The relationship between Flow and AI in video games has also evolved over time, and today there are a number of different ways in which AI can be used to enhance the experience of playing video games. Applying design technics regarding AI to achieve flow in video games can create an engaging and enjoyable experience for the player that is not possible with traditional gaming mechanics. This essay will discuss some of the ways that Artificial Intelligence can be implemented into video games to enable 'flow', and the benefits of using artificial intelligence as a tool for game design. Flow is a concept that has been used in many different fields, but it can be best understood through its application to video games. Flow is a state of being that allows the player to be immersed in the game and lose themselves in its world. This state of mind is often associated with activities that are enjoyable and challenging at the same time. In the past, many games were designed to be fun and entertaining, but that approach did not provide much opportunity for improvement or skill acquisition. With the introduction of more immersive games and virtual reality gaming systems, game designers have incorporated ways that players can become engaged with their environments and achieve a sense of mastery over those worlds. AI can play a critical role in this process by providing the player with challenging and interesting tasks that can help them to achieve a state of flow. In combination with well-designed gameplay elements, AI can be used to enhance the player's experience in a way that is not possible with older forms of video games. AI has been used in many different ways to enhance the experience of playing video games. One of the most common applications of AI in video games is the widespread use of computer-controlled enemies in battle-based games. These enemies may be controlled through the use of pre-programmed behaviors or by the application of complex algorithms based on the behavior of the player character in the game. as game design is more of an iteration on past understanding, the evolution of NPC design has changed radically in just the last 5 years. in the days of pc gaming in the 90s and early 2000s, many of the top-selling fps titles had fairly similar combat systems that featured man vs machine-type gameplay that required you to be constantly on the lookout for the enemy and dodge incoming fire. Wang, K., Sun, W., & Du, Q.'s (2019) "A cooperative game for automated learning of elastoplasticity knowledge graphs and models with AI-guided experimentation." uses a data-driven understanding of AI to analyze how different types of AI can be used to guide video game play to improve learning and skill acquisition specifically when regarding human players. Games like "Deus ex-human revolution" often provide strong examples of this type of game design when highly evolved NPCs that react in real-time to your interactions to progress through a given story provide a compelling experience that encourages repeat playthroughs and engagement with the game. When the system is constantly adapting and changing based on your actions and the way that you interact with the environment, it can lead to an experience that is completely engaging and helps to keep you immersed in the gameplay for long periods of time. There are many different ways that developers can use AI to add additional layers of complexity to their games that can greatly impact the way that a player interacts with the game. AI can also be used to add an experiential element to games that can enhance the level of immersion that the player experiences while they are playing the game. Current Research There is a lot of current research being done in the area of Flow and AI. Whether it's learning to focus attention, manage resources, work with others, or improve their general problem-solving skills, it appears that players can apply the skills that they learn through video games to other aspects of their lives. Additional research suggests that players can learn to work more efficiently and to set goals for themselves to achieve specific goals while playing a video game. This type of learning can be beneficial in many different areas of life, and it is interesting to see how video games are teaching us new ways to improve our skills and be more effective in our day-to-day lives. Christian Arzate Cruz and Jorge

Adolfo Ramirez Uresti’s player-centered game AI from a flow perspective: Towards a better understanding of past trends and future directions (2017) looked at how the concepts of flow and immersion relate to the design of video games and the use of AI in those environments. They argue that developers should focus on creating games that allow players to achieve a state of flow in order to create an engaging and immersive experience. They also point out that in order to achieve this state, the game should be challenging enough to engage the player but not so difficult that it becomes frustrating or stressful. With the new advances to unreal engine 5, it seems possible to create games that have realistic combat environments paired with intelligent AI that can provide the player with opportunities for skill development and growth. These games could allow players to hone their skills and be better able to tackle challenging situations in the real world. Time Speeds Up in Flow States When Playing Video Games (Wittmann, 2021) explains how the use of AI can help to speed up the pace of a flow state in video games. Wittmann conducted a study in which he found that the more challenging the game, the faster time felt to players. According to Wittmann, this is because our brains try to come up with solutions to the challenges that we are facing so we are able to keep our stress levels low and keep playing (Wittmann, 2021). With increased AI comes the potential for more improvement and increased feelings of satisfaction, leading players to keep playing in order to experience these feelings of achievement. AI in Video Games: Toward a More Intelligent Game (Harbing Lou, 2017) also discusses how the presence of AI in video games can enhance the learning experience for players. Lou describes how we see examples of the use of AI in every walk of life. For example, when you search on Google, it uses algorithms to sort through all of the information that is on the Internet to help you find what you are searching for. Flow and immersion are two concepts that are very important to game designers who want to create engaging experiences for their players. Flow and Immersion in Video Games: The Aftermath of a Conceptual Challenge (Michailidis, L., Balaguer-Ballester, E., & He, X. 2018) outlines how the concept of flow has been challenged by the advancement of technology. Flow is a state of intense focus and concentration that is usually associated with tasks that are enjoyable. Immersion is the degree to which a person becomes absorbed in an experience. together, flow and immersion are two concepts that are used to describe the state of mind that people experience when they are engaged fully in a game. Finally, current research on flow suggests that playing video games can lead to positive changes in players’ cognitive abilities and physical behavior. How AI is Used in Game Development? (THINKML TEAM 2022) explains the benefits of human phycology that can be derived from the use of AI in game development. They discuss some of the ways in which AI can be used to help game developers achieve what they want to achieve. For example, AI can be used to create 3D models of the game world so that developers know exactly what objects should look like and where each object should be placed. The current research for flow mixed with AI development is still at the early stages, but there is evidence that points to the possibility that it could help the player gaming experience in the future. Video games today have come a long way since the days of Pong and Space Invaders. Modern games are packed with amazing graphics and realistic sound effects that make them much more immersive and engaging than they used to be. However, this comes at a cost – most games are so big and complicated that they require a lot of processing power to run. Yunlong Lu and Wenxin Li's (2019) "Techniques and Paradigms in Modern Game AI Systems" explains that this can be a problem for modern game developers who need to create new games and updates on a regular basis. AI development has emerged as a potential solution to this problem, as it can automatically generate events and scenarios based on the game data being fed to it. While the research in this area is still in its early stages, there is already enough evidence to show that AI can indeed be a useful addition to many game systems. Harré, M. S.'s (2022) "What Can Game Theory Tell Us about an AI “Theory of Mind”?" argues that game theory can provide some useful insights into the problems that can arise when AI attempts to simulate human intelligence. Harré describes how game theory can be used to study various types of cooperative and competitive games in which players interact with each other using limited resources, then applies this knowledge to the question of whether AI can behave like a human being in social situations. This data concludes that it is unlikely that an AI will be able to acquire a "theory of mind" any time soon, but that it may be possible to create certain types of programs that mimic certain aspects of human behavior which can be used to stimulate and test theories about how people work in video games. In conclusion, while the current research on AI/flow is still in its very early stages, there is some evidence to suggest that it could be a beneficial addition to the future of video game development. New Model Research for flow mixed with AI development is still in the early stages, and many discoveries are left unproven. Further research is needed to determine the full potential of mixing artificial intelligence with the concept of flow to enhance user

experiences. In future research revolving around the flow state, it is important to consider different types of games and how they can be designed in order to create a more immersive and engaging gaming experience. Even if a game does not directly incorporate AI as part of the gameplay, incorporating elements that create a sense of flow can be just as beneficial. For example, providing players with a sense of purpose can be extremely helpful in stimulating flow. In games like FarmVille, the player is required to constantly monitor and care for their crops in order to earn more money and maximize crop yield. By keeping a close eye on the condition of the crops and providing regular care, players are able to maintain a positive mental state and keep themselves entertained for a long period of time. Enhanced by AI, player engagement can and must be maintained throughout the game in order to achieve flow. Aside from the importance of maintaining player engagement, it is also important to consider the study of AI in order to develop more effective game designs. The study of machine learning gives designers an idea of what they are able to create through game design. With AI development, developers are able to create games that are more intuitive and can be used to create situations that can be enjoyable for players while being relevant to the game. By taking these factors into consideration during the design of a game, it can help developers create games that can easily adapt to user input and feedback while also giving players a sense of control and freedom. Games that incorporate AI typically require a certain level of user involvement in order to create an effective experience. Because the user must train the AI system in order for the program to learn how to respond to actions, the user is able to personalize the experience to what they want to experience while playing the game. Well-made AI in video games is made with a series of goals in mind, the ability to create a system that can learn on its own is key to creating an effective and engaging game. User feedback also plays a large role in using AI in the development of games because it provides developers with information about what aspects of the game users enjoy the most and what they would like to see improved upon in the future. The inclusion of a player's personal input into the design process is important in creating a game that is both intuitive and exciting. Designers should be able to communicate between developers and users in order to improve the user experience of a game over time, thus allowing the user to be able to create a more personalized experience with an AI system. In order to create a well-functioning AI system, it is important that developers have a good understanding of the different AI algorithms that are available and how they can be used to create an engaging game. The use of AI can help make games more accessible for players by creating an experience that is easier to personalize. A new model for research into flow in video games must be presented, as the current models do not take into account how AI can be used regarding the relationship between a player and the story and environment in a game. AI can be used to increase the level of immersion in a game by allowing players to interact with characters and environments in a way that is both intuitive and enjoyable. Developers use techniques like procedural generation to create expansive and dynamic game worlds that create a more natural feel for the game and allow the player to navigate these spaces without being constrained by the limitations of traditional level design. Procedural generation takes advantage of AI by generating structures automatically in response to the actions of the player and can be implemented into various game genres to create a more immersive and explorative gaming experience. While these techniques can be used in single-player games to create unique experiences for each character, they can also be used to create multiplayer games in which each player is able to experience the game in a unique way based on their specific skills and abilities. Video games that use AI are able to provide a more immersive experience because the player is able to interact directly with the game characters and their environments. Madan, C. R.' (2020) "Considerations for Comparing Video Game AI Agents with Humans" attempts to compare data from different studies which highlighted that human players experience a higher level of satisfaction when interacting with real rather than simulated characters. These studies highlight the importance of using real-world data in order to create a more intuitive AI system that can more accurately simulate the interactions that players have with a game world. New research needs to be conducted in order to further understand how players are interacting with games and whether the use of AI systems can lead to higher levels of engagement for players and create more enjoyable gaming experiences. Limitations With changes to AI happening rapidly every day, there is a lack of long-term research into how AI can be used in video games. This lack of research has led to a number of misconceptions about AI in video games, and it is important that developers have a better understanding of how AI can be used in order to create an engaging game. Because the current state of AI is ever-evolving, external validity exists in that the study may have altered results if conducted at a different time. It is also important to consider that the data collected may not be generalizable to all gamers; instead, it is more likely that the results reflect the

experiences of the participants in the study. Internal validity is also an issue, as it is possible that the measures used to assess AI were not sensitive enough to capture its impact on player experience. For example, items may have been placed too close together in order to make it harder to distinguish between them and thus might inadvertently affect the results of the study. In addition, it is important to consider that not all games will benefit from the use of AI. Games that rely heavily on interpretation or creativity by the player are not likely to be improved by AI, as it would automate these tasks. Lastly, Social threats exist when working with AI in video games. For example, if a character in the game makes negative comments about the player's appearance or insults them, the social reaction of the player may be stronger than before. This could have a negative impact on the player's experience, as they may feel put-off or threatened. While not all games that use AI are problematic in this way, it is important to be aware of it in order to address it in other games. In conclusion, the study of AI in video games can be an important tool for developers to use in the creation of their games. New research needs to be conducted in order to further understand how players are interacting with games and whether the use of AI systems can lead to higher levels of engagement for players and create more enjoyable gaming experiences. Using artificial intelligence technology, video games can be enhanced to provide players with experiences that transcend the boundaries of the real world. However, more work needs to be done in this area to better understand how the technology works and how best to use it in the design of effective games. Although there are several challenges associated with using AI to create better video game experiences, these challenges can be overcome through a combination of further research and an improved understanding of how technology can be used to improve games. Cao, M.'s (2020) "Merging game theory and control theory in the era of AI and autonomy." shows new information regarding how game theory can be used besides autonomy for controlling behavior in-game and in real life. Just as video games predict real-life behaviors, they can also be used to promote real-world good behaviors; for example, certain levels of completion in a game such as Pokemon Go can be used to track a player's walking habits and provide incentives to continue to exercise. As society becomes more technologically advanced, we see more and more applications of AI in our lives and especially in video games. Conclusion The current state of AI in video games is limited by research, but this lack of knowledge has led to a flawed understanding of its capabilities and limitations. However, as the field grows and progresses, it is likely that more research will be conducted that will improve our understanding of AI in video games. By studying the way that AI has been used in other contexts, such as music and language, researchers may be able to draw lessons from these domains to inform the use of AI in gaming. The ultimate goal of this field is to improve the player experience by inducing the flow state at a deeper level, which may be possible through the use of AI. The indie capstone track focuses on the study of game design, a complex and ever-evolving field. Video games already offer compelling and complex worlds; with the help of AI, players can immerse themselves in these worlds even further and be more mesmerized than ever before. Wang, L., Huang, W., Li, Y., Evans, J., & He, S.'s (2020) "Multi-AI competing and winning against humans in iterated Rock-Paper-Scissors game." provides a great understanding of simple AI adaptability when used in simple games of rock, paper, scissors. From this study, we see that AI was able to win about half the time, outperforming human players in several cases. Games are an exciting and unique testing ground for artificial intelligence because of their adaptability to their environment and their vast potential for creative expression. In conclusion, the current state of AI in video games is limited, but the future looks bright for those interested in its development.

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